

- 1 Consider a ream (500 sheets) of A4 papers (70 or 80 gsm).
 Thickness of 1 ream ≈ 5 cm, so thickness of one piece 0.01 cm ≈ 0.1 mm.
 Area of A4 paper $\approx 200 \times 300 = 60,000$ mm² = 0.06 m²
- (a) (i) $0.05 - 0.15$ mm A1
 (ii) $4 - 5$ g A1
- (b) time = $\frac{0.15 \times 10^{12}}{3.00 \times 10^8}$ C1
 $= 500$ s = 8.3 min A1
- (c) (i) SI units for T : s, R : m and M : kg (or seen in formula) C1
 $K = \frac{T^2 M}{R^3}$ units of $K = \frac{s^2 \text{kg}}{\text{m}^3}$ A1
- (ii) $K = \frac{(86400)^2 (6 \times 10^{24})}{(4.23 \times 10^7)^3} = 5.918 \times 10^{11}$ C1
 $\frac{\Delta K}{K} = 2 \frac{0.5}{100} + 3 \frac{1}{100} + \frac{2}{100} = 0.06$ C1
 $\Delta K = 0.355 \times 10^{11}$
 $K = (5.9 \pm 0.4) \times 10^{11}$ (SI units) A1
 (incorrect % value, then max 1 mark)
- OR, $K_{\max} = 6.283 \times 10^{11}$, $\Delta K = K_{\max} - K = 0.365 \times 10^{11}$
 $K_{\min} = 5.57 \times 10^{11}$, $\Delta K = \frac{1}{2} (K_{\max} - K_{\min}) = 0.355 \times 10^{11}$
- (iii) R , as this has the largest fractional uncertainty. B1

- 2 (a) 15 m s^{-1} A1